Evaluating the Quality of Explanations under a Cognitive Feature Budget

Complexity-Calibrated Local Concordance for compact local explanations

Explainable and Trustworthy AI

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Problem

From full fidelity to displayed fidelity

- Local explainers often produce one contribution per feature.
- Standard fidelity can evaluate the complete additive reconstruction.
- Real interfaces usually display only a small number of features.
- A faithful full explanation can become misleading after truncation.

Core question

How much model fidelity is preserved when only the top-K contributions are shown?

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Contribution

Compact faithfulness under a cognitive budget

- We introduce Complexity-Calibrated Local Concordance.
- CCC measures reconstruction error after enforcing a feature budget $K \in \{5, 6, 7, 8, 9\}$.
- Lower CCC means the displayed explanation better reconstructs the black-box probability.
- The metric complements full MSE, sufficiency, comprehensiveness, and random-K baselines.

CCC Definition

Budget-aware additive reconstruction

$$g_K(x) = w_0 + \sum_{i \in S_K(x)} \phi_i(x)$$

$$\mathrm{CCC}_K = \frac{1}{N} \sum_{j=1}^{N} \left(f(x^{(j)}) - g_K(x^{(j)}) \right)^2$$

- $f(x)$ is the black-box positive-class probability.
- w_0 is the explainer intercept.
- $\phi_i(x)$ is the additive feature contribution.
- $S_K(x)$ keeps the K largest absolute contributions.

Common Explainer Format

Normalizing LIME, SHAP, and MAPLE

Explainer	Additive contribution used by CCC
SHAP	Shapley values are already additive contributions.
LIME	Local coefficients are multiplied by the interpretable representation: $\phi_i = \beta_i z_i(x)$.
MAPLE	Local coefficients are multiplied by the feature value: $\phi_i = \beta_i x_i$.

Assumption

All explainers are evaluated through the same form: $g(x) = w_0 + \sum_i \phi_i(x)$.

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Experimental Setup

Datasets, models, explainers, and budgets

Datasets

- UCI Adult
- UCI Breast Cancer Wisconsin Diagnostic

Explainers

- LIME
- SHAP KernelExplainer
- MAPLE

Classifiers

- XGBoost
- Neural network

Evaluation

- $K = 5, 6, 7, 8, 9$
- Seeds 42, 123, 2026

Benchmark Pipeline

Reproducible local explanation evaluation

1. Load and scale the dataset.
2. Train each configured black-box classifier.
3. Generate local explanations for each model and explainer pair.
4. Convert explanations into additive contributions.
5. Compute CCC, full MSE, random-K, sufficiency, and comprehensiveness.

Implementation

The orchestration is implemented in `core/test_framework.py`; metrics are implemented in `core/metrics.py`.

Model Accuracy

Accuracy alone does not select the best explainer

Dataset	Model	Mean accuracy
Adult	XGBoost	0.874
Adult	Neural network	0.823
Breast Cancer	XGBoost	0.968
Breast Cancer	Neural network	0.977

- High model accuracy does not imply compact explanations.
- CCC measures the displayed explanation, not the classifier alone.

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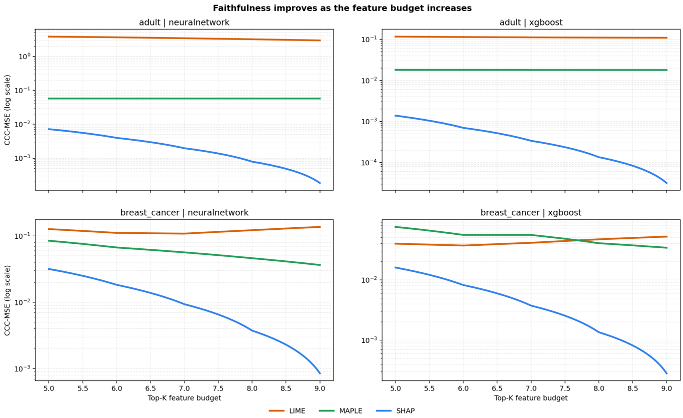
CCC Curves

Lower curves mean more faithful compact explanations



Representative CCC Values

Lower is better

Dataset/model	Explainer	$K = 5$	$K = 6$	$K = 7$	$K = 8$	$K = 9$
Adult/NN	SHAP	0.0072	0.0040	0.0020	0.0008	0.0002
Adult/NN	MAPLE	0.0574	0.0574	0.0574	0.0574	0.0574
Adult/NN	LIME	3.8262	3.6462	3.4261	3.1941	2.9571
Adult/XGB	SHAP	0.0014	0.0007	0.0003	0.0001	0.0000
Adult/XGB	MAPLE	0.0181	0.0180	0.0179	0.0179	0.0179
Adult/XGB	LIME	0.1168	0.1139	0.1119	0.1104	0.1095
Breast/NN	SHAP	0.0318	0.0183	0.0094	0.0037	0.0009
Breast/NN	MAPLE	0.0847	0.0670	0.0565	0.0462	0.0365
Breast/NN	LIME	0.1269	0.1111	0.1084	0.1220	0.1368
Breast/XGB	SHAP	0.0160	0.0082	0.0037	0.0014	0.0003
Breast/XGB	MAPLE	0.0755	0.0556	0.0555	0.0405	0.0341
Breast/XGB	LIME	0.0398	0.0370	0.0412	0.0470	0.0521

Explainer Patterns

What the CCC curves reveal

SHAP

Lowest CCC overall. Error decreases quickly as K grows, but small budgets can still lose fidelity.

MAPLE

Intermediate behavior. Some settings improve with K; Adult is almost flat, suggesting weak top-K concentration.

LIME

Most unstable. On Breast Cancer, CCC is non-monotonic because local surrogate weights do not guarantee additive accuracy.

Why LIME Can Be Non-Monotonic

Adding features can increase squared reconstruction error

- LIME refits a local sparse surrogate and does not enforce local accuracy.
- Feature selection is not guaranteed to be nested across values of K.
- Breast Cancer features are highly correlated, so added terms can change the reconstruction direction.
- Ranking by absolute contribution can add terms whose signs increase $|f(x) - g_K(x)|$.

Example

Breast/NN LIME improves from $K = 5$ to $K = 7$ but worsens at $K = 8$ and $K = 9$: 0.1269, 0.1111, 0.1084, 0.1220, 0.1368.

CCC Versus Full MSE

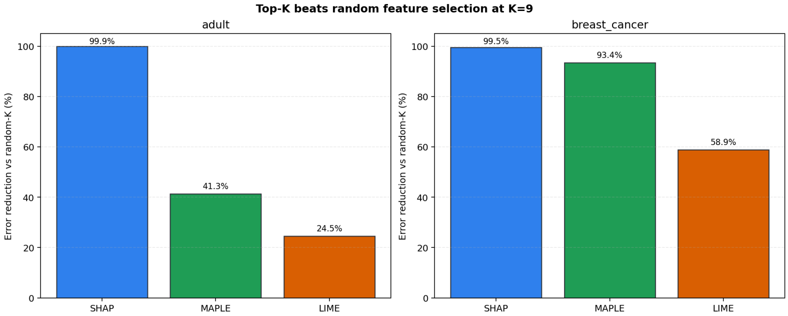
Full fidelity is not compact fidelity

K	5	6	7	8	9
full MSE	0.0	0.0	0.0	0.0	0.0
CCC	0.0160	0.0082	0.0037	0.0014	0.0003

- Example: SHAP on Breast Cancer/XGBoost.
- Full additive reconstruction is exact.
- The displayed top-K explanation still has measurable error.

Top-K Versus Random-K

The ranking should beat arbitrary feature subsets



Metric Comparison

CCC answers a different evaluation question

Metric	Relation to CCC
Full MSE	Evaluates all-feature reconstruction; CCC evaluates after top-K truncation.
Sufficiency	Keeps selected input features under a masking baseline; CCC evaluates additive concordance directly.
Comprehensiveness	Measures prediction change after removing selected features; CCC asks whether displayed additive terms reconstruct $f(x)$.
Infidelity or sensitivity	Measures stability under input perturbations; CCC adds an explicit cognitive feature budget.
Conciseness	Measures explanation size alone; CCC combines compactness with predictive faithfulness.

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4 Takeaways

What CCC Adds

A practical lens for human-sized explanations

- CCC separates model accuracy, full explanation fidelity, and compact explanation fidelity.
- It exposes how quickly the top-ranked features recover the model prediction.
- It can be summarized as a curve, an area under the CCC curve, or a minimum faithful K.
- It is useful as an offline benchmark and as a runtime warning for deployed explanation tools.

Limitations and Extensions

Where the metric should be pushed next

Limitations

- Designed for additive explanations.
- Depends on probability scale and calibration.
- Fixed K is only a proxy for cognitive complexity.
- Current benchmark covers two datasets and two model families.

Extensions

- Add datasets and black-box models.
- Choose the smallest K below an error threshold.
- Integrate CCC into explanation dashboards.

Conclusion

Compact faithfulness is a distinct target

- Full explanation fidelity does not guarantee faithful displayed explanations.
- CCC evaluates the fidelity of the explanation that a user actually sees.
- In this benchmark, SHAP gives the strongest compact reconstructions, MAPLE is intermediate, and LIME is more sensitive to K and local instability.
- Explainer selection should combine CCC with perturbation metrics when practical decisions depend on both reconstruction and input-masking behavior.

Evaluating the Quality of Explanations under a Cognitive Feature Budget

Thank you for listening!

Any questions?